Proton Radius Puzzle and Lattice QCD

Seminar II – Nuclear and Particle Physics

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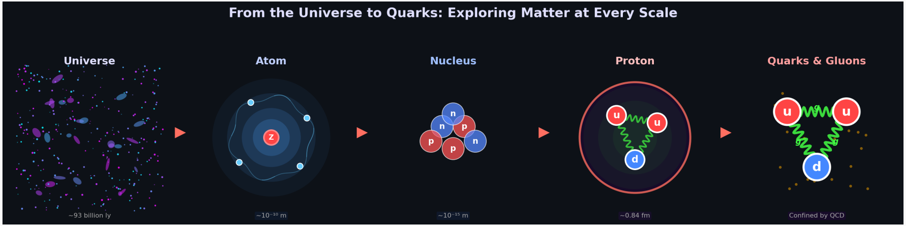
Outline

The Proton and Its Significance

Motivation: From the Universe to the Proton

The central goal of physics: understand the matter of the universe and the laws governing its motion.



- Ordinary visible matter contains quarks and leptons interacting through gauge fields
- The **proton**: quarks + gluons — How big is it?

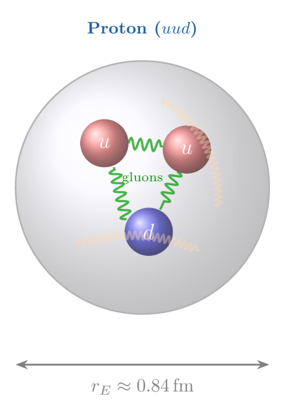
Hofstadter, Rev. Mod. Phys. **28**, 214 (1956) — Nobel Prize 1961.

Why Is the Proton So Important?

Why proton?

1. Ordinary matter: lightest baryon
2. Internal structure: charge, mass, spin
3. QCD benchmark: structure from first principles

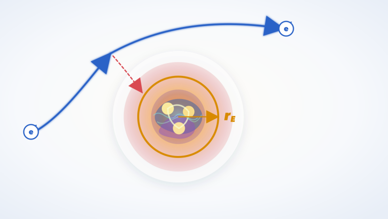
Today: the charge radius r_E^p.

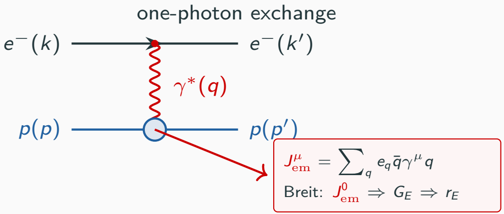


Measuring the Proton Radius

Elastic Scattering Probes r_E

Backup





1. **Measure the differential cross section**

$$\frac{d\sigma}{d\Omega} = \sigma_{\rm Mott}\, \mathcal{S}_p(Q^2, \theta).$$

$\sigma_{\rm Mott}$ is the point-charge QED baseline; $\mathcal{S}_p$ contains the proton structure through G_E and G_M.

2. **Read the electric response at low Q^2**

$$G_E(Q^2) \simeq \int d^3r\, \rho_E(\vec{r})\, e^{i\vec{q}\cdot\vec{r}}$$
$$= 1 - \frac{Q^2}{6}\langle r_E^2\rangle + \mathcal{O}(Q^4).$$

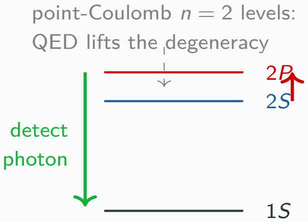
3. **The initial slope defines the charge radius**

$$\langle r_E^2\rangle = -6\, \frac{dG_E}{dQ^2}\bigg|_{Q^2=0}.$$

Hofstadter (1956); Bernauer et al. (A1, 2010); Gao & Vanderhaeghen (2022).

Method 2: Lamb Shift Spectroscopy

Backup



$$\Delta E_{\mathrm{res}} \equiv E(2P) - E(2S)$$

The laser measures ΔE_{res}. Known fine- and hyperfine-structure terms are removed to isolate the Lamb shift.

How is the resonance detected?

$$2S_{\mathrm{meta}} \xrightarrow{\text{laser}} 2P \xrightarrow{\text{photon}} 1S.$$

1. Prepare long-lived $2S$ muonic hydrogen.
2. Scan the $2S \to 2P$ laser frequency.
3. Detect the prompt $2P \to 1S$ photon.

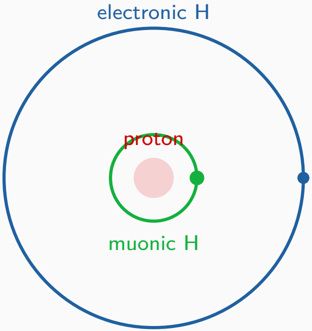
Measured resonance (2010)

$$\Delta E_{\mathrm{res}} = 206.2949(32)\ \mathrm{meV},$$
$$r_E^p = 0.84184(67)\ \mathrm{fm}.$$

Pohl et al., Nature **466**, 213 (2010); Antognini et al., Science **339**, 417 (2013).

Why Muonic Hydrogen Is So Sensitive

Backup



Muon orbit $\sim$ 200 times smaller

Finite proton size shifts an S-state

$$\Delta E_{\mathrm{fs}}(nS) \propto |\psi_{nS}(0)|^2\, r_E^2 \propto m_r^3\, r_E^2.$$

- S-waves reach the proton: $|\psi_{nS}(0)|^2 > 0$.
- P-waves vanish at the origin and provide a nearby reference.
- Since $m_\mu/m_e \simeq 207$, the finite-size sensitivity is enhanced by roughly 10^7.

Pohl et al., Nature **466**, 213 (2010); Antognini et al., Science **339**, 417 (2013).

Proton Radius Puzzle (2010–2026)

Backup

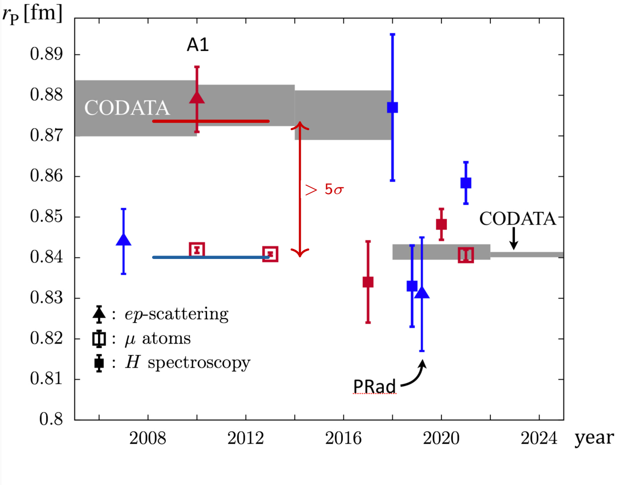


Figure message
Old split:

$$\mu\mathrm{H}: 0.84184(67)\ \mathrm{fm},$$
$$\mathrm{CODATA\text{-}2010}: 0.8775(51)\ \mathrm{fm}.$$

Same proton, different probes: $> 5\sigma$.
This was the puzzle.

Modern checks:
PDG 2024: $r_E^p = 0.8409(4)$ fm;
H $2S$–$6P$ (2026): $0.8406(15)$ fm.

Why puzzle?

$$\langle r_E^2 \rangle = -6\, G_E'(0)$$

Same proton current $J^\mu_{\mathrm{em}} \Rightarrow$ same $G_E'(0)$: a proton property.

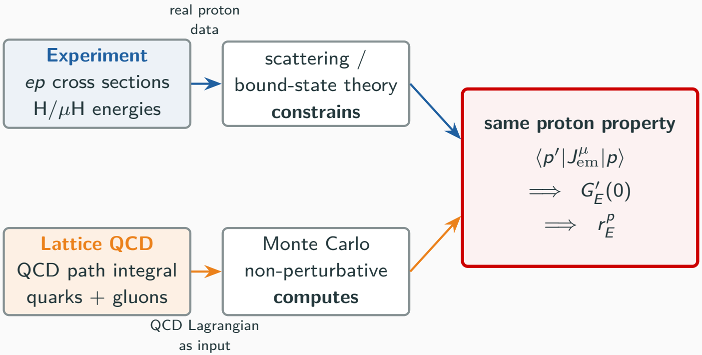
Pohl et al., Nature **466**, 213 (2010); CODATA 2010; PDG 2024; Maisenbacher et al., Nature **650**, 845 (2026).

Lattice QCD

Why Lattice QCD? From Measurement to QCD Prediction

Backup

After experiments: can QCD itself predict the same proton response?



Message: experiments infer from data; lattice QCD computes from QCD; both use the same radius definition.

QCD Observables Are Operator Averages

Backup

A QCD observable is an average over fields

$$\langle O \rangle = \frac{1}{Z_E} \int [D\Phi]\, O[\Phi]\, e^{-S_E[\Phi]}, \quad \Phi = (U, \bar{\psi}, \psi).$$

$$\boxed{t \to -i\tau : \qquad e^{iS_M[\Phi]} \longrightarrow e^{-S_E[\Phi]}}$$

Why Euclidean?

- e^{iS_M} is oscillatory.
- e^{-S_E} behaves like a statistical weight.
- The action tells us which field configurations are likely.

Monte Carlo average

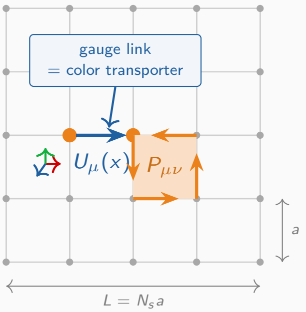
$$\langle f \rangle = \int dx\, f(x) P(x) \simeq \frac{1}{N} \sum_i f(x_i).$$

Draw likely configurations $x_i \sim P(x)$, measure $f(x_i)$, average.

On the lattice: $U_i \sim P[U]$, then $\langle O \rangle \simeq N_{\mathrm{cfg}}^{-1} \sum_i O[U_i]$, with $O = C_2$ or C_3^{μ}.

Discretising QCD: Sites and Links

Why links?



site: quark field $\psi(x)$ link: transporter $U_\mu(x)$

Wilson, PRD **10**, 2445 (1974); Wittig, §5.2.2.

Quarks live on sites

$$\psi(x) \to \Omega(x)\psi(x)$$

$\psi(x)$ is a colour vector at one lattice point.

Gluons live on links

$$U_\mu(x) = \mathcal{P} e^{ig \int_x^{x+a\hat{\mu}} A_\mu(z)\, dz^\mu}$$

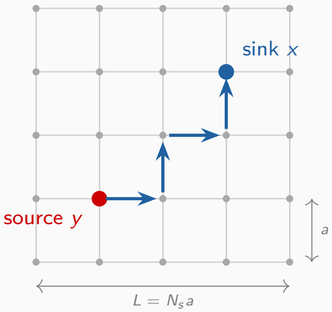
U_μ **parallel-transports colour** from x to $x + a\hat{\mu}$.

$$\bar{\psi}(x)\, U_\mu(x)\, \psi(x + a\hat{\mu}) \quad \text{is gauge invariant.}$$

Plaquette A plaquette $P_{\mu\nu}$ is the smallest closed link loop; summing plaquettes gives $S_g[U]$, with $a \to 0$ recovering $\mathrm{Tr}\, F_{\mu\nu} F_{\mu\nu}$ in Euclidean space.

From Links to Quark Green's Functions

Backup



quark propagator = response from y to x

1. Fix one gauge background U_i

All links are known, so the lattice Dirac operator $D_q[U_i; m_q]$ is known.

2. Solve for the quark propagator

$$\sum_z D_q(x, z; U_i)\, S_q(z, y; U_i) = \delta_{xy}, \qquad S_q = D_q^{-1}.$$

$$\text{point source at } y \quad \longrightarrow \quad \text{response at } x.$$

This response is the quark Green's function.

3. Build proton observables

The proton operator contracts three propagators (S_u, S_d, S_u) into C_2 and C_3^μ; finally average over all U_i.

$$U_i \Longrightarrow D_q[U_i] \Longrightarrow S_q = D_q^{-1} \Longrightarrow C_2, C_3^\mu.$$

Why Euclidean Time? Filtering the Proton

Backup

Proton operator: a *uud* source/sink

$$\chi_p^\alpha(x) = \epsilon^{abc}\left(u^{aT}(x) C\gamma_5 d^b(x)\right) u^{c\alpha}(x).$$

χ_p has proton quantum numbers and overlaps with proton/excited states.

Two-point correlator

$$C_2(t) = \sum_{\vec{x}} \langle \chi_p(\vec{x}, t)\, \bar{\chi}_p(\vec{0}, 0)\rangle.$$

Euclidean time acts as a filter

$$\begin{aligned} C_2(t) &= \langle 0|\chi_p e^{-Ht} \bar{\chi}_p|0\rangle \\ &= \sum_n |A_n|^2 e^{-E_n t} \xrightarrow[t \text{ large}]{} |A_0|^2 e^{-M_N t}. \end{aligned}$$

$\bar{\chi}_p|0\rangle = \sum_n A_n|n\rangle$: not an energy eigenstate.
Large t keeps the lowest state $E_0 = M_N$.

Large t: excited states die out; the slope of $\log C_2(t)$ gives M_N.

From Correlators to the Proton Current

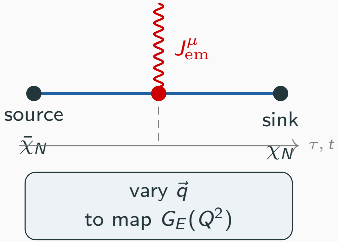
Backup

Insert the electromagnetic current

$$C_3^\mu(t,\tau,\vec{q}) = \langle \chi_N(t)\, J^\mu_{\rm em}(\tau,\vec{q})\, \bar{\chi}_N(0)\rangle.$$

$$0 \ll \tau \ll t \quad \Longrightarrow \quad C_3^\mu\Big|_{\rm ground\ state} \propto \langle p'|J^\mu_{\rm em}|p\rangle.$$

This is the lattice version of a virtual scattering experiment: create a proton, insert the current, destroy the proton.



Then decompose

$$\langle p'|J^\mu_{\rm em}|p\rangle \to F_1(Q^2), F_2(Q^2) \to G_E(Q^2), G_M(Q^2).$$

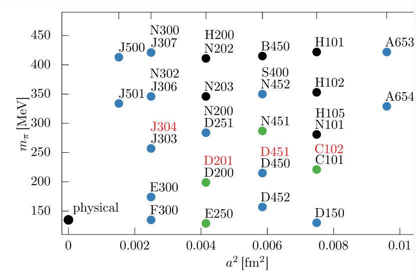
Sachs, Phys. Rev. **126**, 2256 (1962); Djukanovic et al., PRL **132**, 211901 (2024).

Gauge Ensembles at Mainz

Why Many Lattice Ensembles?

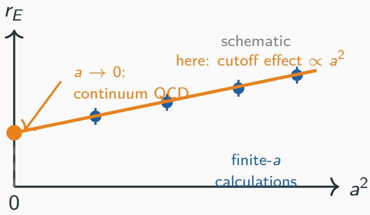
Backup

One ensemble is one finite lattice. The physical result requires several lattice spacings.



CLS varies a, m_π, and the box size L.

RG: change the cutoff while keeping the physics fixed



Changing a changes the cutoff $\Lambda_{\mathrm{UV}} \sim 1/a$. Repeat the same observable at several matched a, then take the intercept at $a = 0$.

Bruno et al., JHEP **02**, 043 (2015); Bali et al., PRD **94**, 074501 (2016).

Computing Form Factors on the Lattice

From QFT to the Lattice: Correlation Functions

Backup

Goal: compute $G_E(Q^2)$ and $G_M(Q^2)$ from lattice QCD.

Two-point function (χ_N: nucleon interpolating operator):

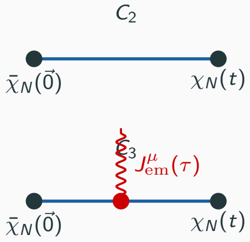
$$C_2(\vec{p},t) = \sum_{\vec{x}} e^{-i\vec{p}\cdot\vec{x}} \langle \chi_N(\vec{x},t)\bar{\chi}_N(\vec{0},0)\rangle$$

$$\xrightarrow{t\ \text{large}} |\mathcal{A}_0|^2 e^{-E_{\vec{p}}t}.$$

Three-point function (nucleon–current–nucleon, with $\vec{p}\,' = 0$):

$$C_3^\mu(\vec{q},t,\tau) = \sum_{\vec{x},\vec{y}} e^{i\vec{q}\cdot\vec{y}}$$

$$\times \langle \chi_N(\vec{x},t)\, J^\mu_{\text{em}}(\vec{y},\tau)\, \bar{\chi}_N(\vec{0},0)\rangle.$$

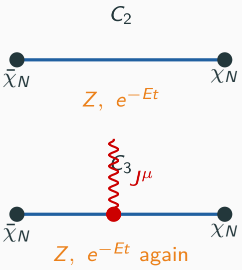


C_2: proton propagation. C_3: insert J^μ_{em} and measure the response.

Djukanovic et al., PRL **132**, 211901 (2024) [arXiv:2309.07491].

Why Take a Ratio?

Backup



Large-time behaviour

$$C_2(t,\vec{p}) \to |Z_p|^2\, e^{-E_p t}.$$

$$C_3^\mu(t,\tau) \to Z_{p'} Z_p^*\, e^{-E_{p'}(t-\tau)} e^{-E_p \tau}\, \langle p'|J^\mu_{\mathrm{em}}|p\rangle.$$

$$R^\mu \sim \frac{C_3^\mu}{C_2\ \mathrm{combinations}} \to \text{known kinematics} \times \langle p'|J^\mu_{\mathrm{em}}|p\rangle.$$

Overlap and propagation factors cancel; the current matrix element remains.

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012).

Extracting Form Factors

Backup

Core idea: the ratio gives a plateau Π_μ; kinematics turns it into G_E and G_M.

Ground-state matrix element

$$R_\mu(\vec{q}, t, \tau) \xrightarrow{0 \ll \tau \ll t} \Pi_\mu(\vec{q}).$$

Plateau = projected ground-state matrix element.

Practical tension: ground-state isolation vs. noise.

Kinematic projection

$$\Pi_\mu(\vec{q}) = K_\mu^E(\vec{q})\, G_E(Q^2) + K_\mu^M(\vec{q})\, G_M(Q^2).$$

Different current components and momenta give enough equations to determine the two Sachs form factors.

Then

- repeat for several Q^2
- fit $G_E(Q^2)$ near zero
- slope $\to$ charge radius

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012).

Quark-Line Disconnected Diagrams

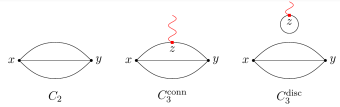
Backup

The current can enter in two ways

Connected
The current hits a quark line joining the proton source and sink.

Disconnected
The current creates a sea-quark loop, correlated with the proton only through the same gluon background.

Numerically noisy and expensive, but required for precision.



C_2, connected C_3, and disconnected C_3.

Djukanovic et al., PRL **132**, 211901 (2024) — includes disconnected diagrams in a precision nucleon-radius calculation.

Electromagnetic Form Factors: The Bridge to the Radius

Backup

Figure message: lattice QCD computes the same $G_E(Q^2)$ curve inferred from scattering; its low-Q^2 slope determines r_E^p.

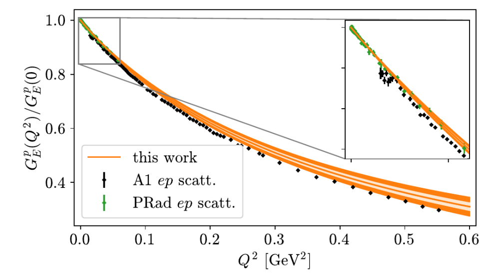
Low-Q^2 slope

$$G_E(Q^2) = 1 - \frac{Q^2}{6}\langle r_E^2\rangle + \cdots$$

$$\boxed{\langle r_E^2\rangle = -6\,\frac{dG_E}{dQ^2}\bigg|_0}$$

Lattice chain

$$C_2, C_3^\mu \to \langle p'|J^\mu|p\rangle$$
$$\to G_E(Q^2) \to r_E.$$



Orange: lattice QCD; green: PRad; black: A1 Mainz *ep* data.

Sachs, Phys. Rev. **126**, 2256 (1962); Djukanovic et al., PRL **132**, 211901 (2024).

Results

Mainz/CLS Proton Charge Radius

Final r_E^p after continuum and physical-point fits.

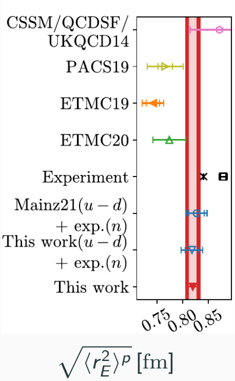
Backup

Djukanovic et al. (2024)

$$r_E^p = 0.820(14)\ \mathrm{fm}$$

Compatible with the modern small-radius picture.

This is a first-principles QCD result after controlled extrapolations.



Djukanovic et al., PRL **132**, 211901 (2024) [arXiv:2309.07491].

Lattice vs. Experiment: Comparison

Lattice QCD

$$r_E^p = 0.820(14)\ \mathrm{fm}$$

PDG 2024

$$r_E^p = 0.8409(4)\ \mathrm{fm}$$

Compatible within about 1.5σ.

Lattice QCD supports the modern small-radius picture, but is not yet at PDG precision.

Lattice: Djukanovic et al., PRL **132**, 211901 (2024); double errors are stat/syst. Exp.: PDG 2024.

Summary and Outlook

Summary: What Did We Learn?

1. **Experiments infer the radius.**

$$\text{cross sections} \to G_E(Q^2), \qquad \text{energy levels} \to r_E^p, \qquad \langle r_E^2 \rangle = -6\, G_E'(0).$$

2. **Modern experiments favour the small-radius picture.**

$$\mu\mathrm{H}: \ 0.84184(67)\ \mathrm{fm}, \qquad \text{PDG 2024}: \ 0.8409(4)\ \mathrm{fm}.$$

3. **Lattice QCD computes the same proton response from QCD.**

$$C_2,\ C_3^\mu \to \langle p' | J^\mu_{\mathrm{em}} | p \rangle \to G_E(Q^2) \to \boxed{r_E^p = 0.820(14)\ \mathrm{fm}}.$$

The Mainz/CLS result is compatible with the modern picture within about 1.5σ.

Outlook: From Consistency to Precision

Lattice QCD

- more configurations / measurements $\Rightarrow$ reduce statistical errors
- finer-a ensembles $\Rightarrow$ controlled $a \to 0$ limit
  (costly: critical slowing down)
- learned / ML samplers $\Rightarrow$ may accelerate gauge-field generation

Goal: first-principles QCD from consistency check to sub-percent precision.

Critical slowing: Schaefer et al., NPB **845**, 93 (2011). ML samplers: Albergo et al. (2019), Kanwar et al. (2020).

Thank you!

Questions?

Backup Roadmap

Puzzle & experiments

- Method tension
- Lamb shift size
- Zemach/Friar

Form factors

- *ep* formula map
- OPE amplitude
- Why Mott factor
- Feynman diagram
- Current form
- *ep* cross section
- Slope as radius
- G_M figure
- Why Sachs?

Lattice details

- Why non-perturbative?
- Why links?
- Why Euclidean?
- Proton operator
- Wick contraction
- Disconnected
- Flavor currents
- Fig. 3 labels
- Ratio method
- Gauge weight
- z-expansion
- Fermion actions

Why Could the Methods Disagree? (Backup)

Literature framing. Pohl et al. (2013) define the puzzle as the significant disagreement between μH spectroscopy and the radius extracted from electronic H / *ep* scattering. Later reviews describe the response as a program of checking theory, questioning experimental systematics, and designing new measurements.

Where could the tension enter?

- ***ep* scattering**: low-$Q^2 \to 0$ extrapolation, absolute normalisation, radiative corrections, two-photon exchange.
- **H / μH spectroscopy**: bound-state QED, R_∞–r_E^p correlation, proton polarizability / two-photon exchange.
- **More radical**: lepton-universality violation or muon-specific new interactions; interesting but highly constrained.

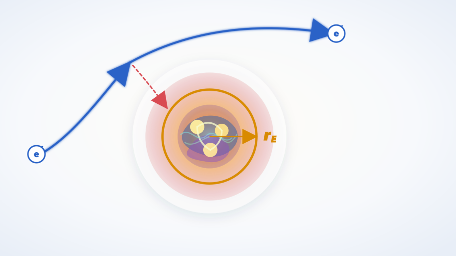
How did people try to improve it?

- **PRad**: very low Q^2, windowless target, magnetic-spectrometer-free normalisation.
- **New H spectroscopy**: different transitions to separate R_∞ and r_E^p; stress-test bound-state QED.
- **MUSE, MAGIX, PRad-II**: new e/μ scattering and lower Q^2 data to isolate systematics.

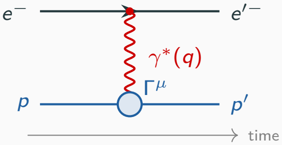
Modern wording: refined *ep* and H measurements moved toward the μH value, but the issue is best viewed as a precision-consistency test of scattering, spectroscopy, QED theory, and possible lepton universality.

Pohl et al., Ann. Rev. Nucl. Part. Sci. **63**, 175 (2013); Karr et al., Nat. Rev. Phys. **2**, 601 (2020); Xiong et al., Nature **575**, 147 (2019); Carlson & Freid, PRD **92**, 095024 (2015); Maisenbacher et al., Nature **650**, 845 (2026).

Elastic Scattering Formula Map (Backup)



one-photon exchange



$$\frac{d\sigma}{d\Omega} = \sigma_{\text{Mott}}\, \mathcal{S}_p(Q^2, \theta)$$

$$\sigma_{\text{Mott}} = \frac{\alpha^2 \cos^2(\theta/2)}{4E^2 \sin^4(\theta/2)} \frac{E'}{E}$$

point-charge QED baseline

$$\mathcal{S}_p = (G_E^2 + \tau G_M^2)/(1+\tau) + 2\tau G_M^2 \tan^2\frac{\theta}{2}$$

proton structure

$$J^{\mu}_{\text{em}} = \sum_q e_q\, \bar{q}\gamma^{\mu} q$$

EM current operator

$$\langle p'|J^{\mu}_{\text{em}}|p\rangle = \bar{u}(p')\left[\gamma^{\mu} F_1 + \frac{i\sigma^{\mu\nu} q_{\nu}}{2M_p} F_2\right] u(p)$$

Dirac/Pauli form factors

$$G_E = F_1 - \tau F_2, \qquad G_M = F_1 + F_2$$

Sachs form factors

$$\langle r_E^2 \rangle = -6 \left.\frac{dG_E}{dQ^2}\right|_{Q^2=0}$$

charge radius

Hofstadter, Rev. Mod. Phys. **28**, 214 (1956); Bernauer et al. (A1), PRL **105**, 242001 (2010); Gao & Vanderhaeghen, RMP **94**, 015002 (2022).

Where Does the Factorised Amplitude Come From? (Backup)

Start from the path integral / S-matrix expansion Local $U(1)_{\rm em}$ gauge invariance gives minimal coupling: the photon field A_μ couples to the conserved electromagnetic current.

$$Z = \int [D\Phi]\, \exp\left\{ iS_0 + i\int d^4x\, \mathcal{L}_{\rm int}(x) \right\}, \qquad \mathcal{L}_{\rm int} = -e\, A_\mu\, j_e^\mu - e\, A_\mu\, J_{\rm em}^\mu .$$

$$j_e^\mu = \bar{e}\gamma^\mu e, \qquad J_{\rm em}^\mu = \sum_q Q_q\, \bar{q}\gamma^\mu q, \quad Q_u = \frac{2}{3},\ Q_d = Q_s = -\frac{1}{3}.$$

$$e^{i\int \mathcal{L}_{\rm int}} = 1 + i\int d^4x\, \mathcal{L}_{\rm int}(x) + \frac{i^2}{2}\int d^4x\, d^4y\, T\{\mathcal{L}_{\rm int}(x)\mathcal{L}_{\rm int}(y)\} + \cdots .$$

Lowest connected $ep \to ep$ term

$$\frac{i^2}{2}\, 2e^2 \int d^4x\, d^4y\, T\{A_\mu(x) j_e^\mu(x)\, A_\nu(y) J_{\rm em}^\nu(y)\}.$$

$$\langle T\, A_\mu(x) A_\nu(y)\rangle_0 = D_{\mu\nu}(x-y) \quad \Longrightarrow \quad \text{one photon line between the two currents.}$$

$$\langle e(k')|j_e^\mu|e(k)\rangle = \underbrace{\bar{u}(k')\gamma^\mu u(k)}_{\ell^\mu \text{ known}} \qquad\qquad \langle p'|J_{\rm em}^\nu|p\rangle = \underbrace{H^\nu}_{\text{proton response}}$$

$$i\mathcal{M} = (-ie)^2 \ell_\mu \frac{-ig^{\mu\nu}}{q^2} H_\nu \quad \Rightarrow \quad \mathcal{M} \simeq \frac{e^2}{Q^2} \underbrace{\ell_\mu}_{known} \underbrace{H^\mu}_{proton}, \qquad Q^2 = -q^2.$$

Why a Mott Factor Times Proton Structure? (Backup)

One-photon exchange separates the known lepton current from the proton current

$$\mathcal{M} = \frac{e^2}{Q^2}\,\underbrace{\bar{u}(k')\gamma_\mu u(k)}_{\ell_\mu\ \text{known}}\,\underbrace{\langle p'|J^\mu_{\rm em}|p\rangle}_{H^\mu\ \text{proton}}\,.$$

$$\overline{|\mathcal{M}|^2} = \frac{e^4}{Q^4}\,L_{\mu\nu}\,W^{\mu\nu}, \qquad L_{\mu\nu} = \ell_\mu \ell_\nu^*, \quad W^{\mu\nu} = H^\mu H^{\nu *}.$$

Point-charge baseline

$$\left(\frac{d\sigma}{d\Omega}\right)_{\rm point} \equiv \sigma_{\rm Mott}.$$

known QED lepton current, photon propagator, and elastic kinematics.

Real proton response

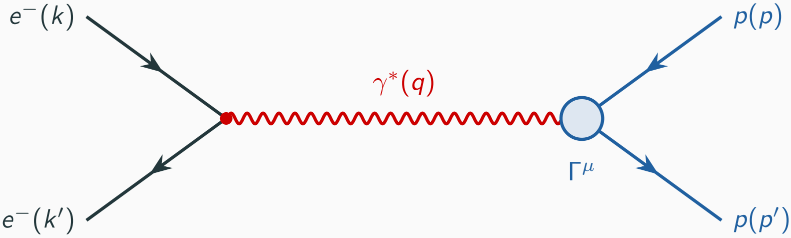
$$W^{\mu\nu}_{\rm elastic} \Longrightarrow G_E(Q^2),\ G_M(Q^2).$$

covariance, current conservation, and spin-$\frac{1}{2}$ leave two elastic responses.

$$\frac{d\sigma}{d\Omega} = \underbrace{\sigma_{\rm Mott}}_{\text{known point-charge QED}}\quad \underbrace{\mathcal{S}_p(Q^2,\theta)}_{\text{dimensionless proton structure}}\,.$$

This is a convenient factorisation of the one-photon-exchange result, not an extra dynamical assumption.

Form Factors from the Feynman Diagram (Backup)



$$\langle p'|J^\mu_{\mathrm{em}}(0)|p\rangle = \bar{u}(p')\left[\gamma^\mu F_1(Q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2M_N}F_2(Q^2)\right]u(p), \qquad Q^2 = -q^2.$$

Peskin & Schroeder, *An Introduction to Quantum Field Theory*, Ch. 6; Sachs, Phys. Rev. **126**, 2256 (1962).

Why This Proton Current Form? I (Backup)

Start from the unknown proton vertex

$$\langle p'|J^{\mu}_{\mathrm{em}}(0)|p\rangle = \bar{u}(p')\,\Gamma^{\mu}(P,q)\,u(p), \qquad P = p' + p, \quad q = p' - p, \quad Q^2 = -q^2.$$

1. Lorentz covariance

$$\Gamma^{\mu} = a_1\gamma^{\mu} + a_2\frac{P^{\mu}}{2M_p} + a_3\frac{q^{\mu}}{2M_p} + a_4\frac{i\sigma^{\mu\nu}q_{\nu}}{2M_p}.$$

- Γ^{μ} must transform as a vector.
- $a_i = a_i(Q^2)$: the only invariant left is q^2.
- $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^{\mu},\gamma^{\nu}]$ carries spin; $\sigma^{\mu\nu}q_{\nu}$ gives the Pauli/magnetic response.
- Parity forbids axial structures such as $\gamma^{\mu}\gamma_5$.

Why This Proton Current Form? II (Backup)

2. Current conservation

$$q_\mu \langle p' | J^\mu_{\mathrm{em}} | p \rangle = 0.$$

$$q \cdot P = p'^2 - p^2 = 0,$$

$$q_\mu \sigma^{\mu\nu} q_\nu = 0 \quad (\sigma^{\mu\nu} = -\sigma^{\nu\mu}).$$

$$\begin{aligned} \bar{u}(p')\, \gamma \cdot q \, u(p) &= \bar{u}(p')(\gamma \cdot p' - \gamma \cdot p) u(p) \\ &= 0. \end{aligned}$$

So the q^μ term is removed: $a_3 = 0$.

Why This Proton Current Form? III (Backup)

3. Gordon identity removes the P^μ basis

$$\bar{u}(p')\frac{P^\mu}{2M_p}u(p) = \bar{u}(p')\left[\gamma^\mu - \frac{i\sigma^{\mu\nu}q_\nu}{2M_p}\right]u(p).$$

Therefore the elastic proton current needs two scalar functions

$$\boxed{\langle p'|J^\mu_{\rm em}|p\rangle = \bar{u}(p')\left[\gamma^\mu F_1(Q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2M_p}F_2(Q^2)\right]u(p)}$$

$$F_1^p(0) = 1, \quad F_2^p(0) = \kappa_p = 1.793; \qquad F_1^n(0) = 0, \quad F_2^n(0) = \kappa_n = -1.913.$$

F_1, F_2 are the two scalar functions containing QCD dynamics.

From One-Photon Exchange to the Radius I (Backup)

Insert the proton current into the one-photon exchange amplitude

$$i\mathcal{M} = (-ie)^2\, \bar{u}_e(k')\gamma_\mu u_e(k)\, \frac{-ig^{\mu\nu}}{q^2}\, \bar{u}_p(p')\Gamma_\nu(q)u_p(p), \qquad q = k - k' = p' - p.$$

$$\Gamma^\nu(q) = \gamma^\nu F_1(Q^2) + \frac{i\sigma^{\nu\alpha}q_\alpha}{2M_p}F_2(Q^2), \qquad Q^2 = -q^2, \quad \tau = \frac{Q^2}{4M_p^2}.$$

Sachs variables

$$\begin{aligned} G_E &= F_1 - \tau F_2, \\ G_M &= F_1 + F_2, \\ \epsilon &= \left[1 + 2(1+\tau)\tan^2\frac{\theta_e}{2}\right]^{-1}. \end{aligned}$$

From One-Photon Exchange to the Radius II (Backup)

Mott baseline and reduced cross section

$$\sigma_{\text{Mott}} \equiv \left(\frac{d\sigma}{d\Omega}\right)_{\text{point}} = \frac{\alpha^2 \cos^2(\theta_e/2)}{4E^2 \sin^4(\theta_e/2)} \frac{E'}{E},$$

$$\sigma_R \equiv \frac{\epsilon(1+\tau)}{\sigma_{\text{Mott}}} \frac{d\sigma}{d\Omega} = \epsilon G_E^2 + \tau G_M^2.$$

Rosenbluth separation

$$\text{fixed } Q^2: \quad \sigma_R(\epsilon) = \underbrace{G_E^2}_{\text{slope}} \epsilon + \underbrace{\tau G_M^2}_{\text{intercept}}.$$

Vary E_{beam} and θ_e to scan ϵ at fixed Q^2.

Charge radius

$$G_E(Q^2) = 1 - \frac{1}{6}\langle r_E^2\rangle Q^2 + \mathcal{O}(Q^4),$$

$$\langle r_E^2\rangle = -6 \left.\frac{dG_E}{dQ^2}\right|_{Q^2=0}.$$

Why the G_E Slope Is a Radius (Backup)

Breit-frame intuition

$$G_E(Q^2) \simeq \int d^3r\, \rho_E(\vec r) e^{i\vec q\cdot\vec r}, \qquad \int d^3r\, \rho_E(\vec r) = 1.$$

- $Q^2 = \vec q^{\,2}$ in the Breit frame.
- $G_E(0) = 1$: total proton charge.
- Larger charge radius $\Rightarrow$ faster falloff with Q^2.

Small-q expansion

$$e^{i\vec q\cdot\vec r} = 1 + i\vec q\cdot\vec r - \frac{1}{2}(\vec q\cdot\vec r)^2 + \cdots.$$

$$\int \rho_E\, \vec q\cdot\vec r = 0, \qquad \int \rho_E (\vec q\cdot\vec r)^2 = \frac{\vec q^{\,2}}{3}\langle r_E^2\rangle.$$

$1/6 = (1/2)\times(1/3)$: Taylor expansion $\times$ isotropic average.

$$G_E(Q^2) = 1 - \frac{Q^2}{6}\langle r_E^2\rangle + \mathcal{O}(Q^4).$$

$$\boxed{\langle r_E^2\rangle = -6\,\frac{d\,G_E}{dQ^2}\bigg|_{Q^2=0}}$$

This is the nonrelativistic/Breit-frame picture; the exact relativistic definition is the current matrix element.

Magnetic Form Factor G_M (Backup)

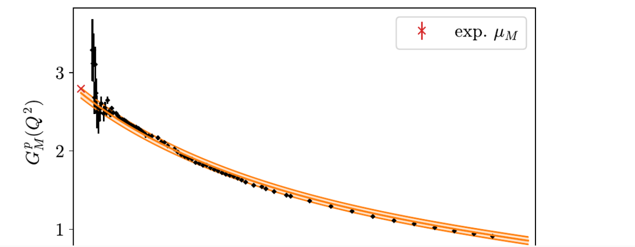
Same lattice calculation also gives G_M, but it is not needed for the charge radius.

Magnetic Sachs form factor

$$G_M = F_1 + F_2.$$

$$\langle r_M^2 \rangle = -\frac{6}{G_M(0)} \frac{dG_M}{dQ^2}\bigg|_{Q^2=0}.$$

- G_E: charge response $\to r_E$
- G_M: magnetic response $\to r_M$
- Main talk follows G_E only



Djukanovic et al. (2024), PRL Fig. 2: $G_M^p(Q^2)$ panel.

Why Sachs G_E, G_M? I (Backup)

Step 1: F_1, F_2 **come from symmetry, not from "electric vs magnetic"**

$$\langle p'|J^\mu_{\mathrm{em}}|p\rangle = \bar{u}(p')\left[\gamma^\mu F_1(Q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2M_p}F_2(Q^2)\right]u(p).$$

- This is the most general spin-$\frac{1}{2}$ current allowed by Lorentz covariance and current conservation.
- F_1 and F_2 are therefore **covariant form factors**.
- At finite Q^2, they do not separately equal "charge distribution" and "magnetisation distribution".

Step 2: choose a frame where charge and current are easy to read

$$\text{Breit frame:} \qquad q^\mu = (0, \mathbf{q}), \quad p^\mu = (E, -\mathbf{q}/2), \quad p'^\mu = (E, \mathbf{q}/2), \quad Q^2 = \mathbf{q}^2.$$

- No energy transfer: $q^0 = 0$.
- J^0 behaves like a charge-density response.
- $\mathbf{J}$ behaves like a current/magnetic response.

Why Sachs G_E, G_M? II (Backup)

Step 3: in the Breit frame, the current components pick special combinations

$$\langle p'|J^0_{\mathrm{em}}|p\rangle_{\mathrm{B}} \;\propto\; \chi'^{\dagger}\chi\,\underbrace{(F_1 - \tau F_2)}_{G_E(Q^2)}, \qquad \tau = \frac{Q^2}{4M_p^2}.$$

$$\langle p'|\mathbf{J}_{\mathrm{em}}|p\rangle_{\mathrm{B}} \;\propto\; i\,\chi'^{\dagger}\frac{\boldsymbol{\sigma}\times\mathbf{q}}{2M_p}\chi\,\underbrace{(F_1 + F_2)}_{G_M(Q^2)}.$$

$$\boxed{G_E = F_1 - \tau F_2}, \qquad \boxed{G_M = F_1 + F_2}.$$

Why "electric"?

$$J^0 \quad\longrightarrow\quad G_E(Q^2)$$

$$G_E(Q^2) \sim \int d^3r\, e^{i\mathbf{q}\cdot\mathbf{r}}\rho_E(\mathbf{r}) \quad \text{(nonrelativistic picture).}$$

$$\langle r_E^2\rangle = -6\,\frac{dG_E}{dQ^2}\bigg|_{Q^2=0}.$$

Why "magnetic"?

$$\mathbf{J} \quad\longrightarrow\quad G_M(Q^2)$$

$G_M(0) = \mu_p$, spatial current encodes magnetisation.

$$\langle r_M^2\rangle = -\frac{6}{G_M(0)}\,\frac{dG_M}{dQ^2}\bigg|_{Q^2=0}.$$

Why $\Delta E_{\mathrm{fs}} \propto |\psi(0)|^2 r_E^2$? (Backup)

Use the same G_E that scattering measures

Slope derivation

$$G_E(Q^2) = 1 - \frac{Q^2}{6}\langle r_E^2\rangle + \mathcal{O}(Q^4).$$

$$V(\mathbf{q}) = -\frac{4\pi Z\alpha}{\mathbf{q}^2} G_E(\mathbf{q}^2) = -\frac{4\pi Z\alpha}{\mathbf{q}^2} + \frac{2\pi Z\alpha}{3}\langle r_E^2\rangle + \cdots .$$

$$\delta V(\mathbf{r}) = \frac{2\pi Z\alpha}{3}\langle r_E^2\rangle\, \delta^{(3)}(\mathbf{r}).$$

First-order perturbation theory

$$\Delta E^{\mathrm{fs}}_{n\ell} = \langle n\ell m|\delta V|n\ell m\rangle = \frac{2\pi Z\alpha}{3}\langle r_E^2\rangle |\psi_{n\ell m}(0)|^2.$$

$$|\psi_{nS}(0)|^2 = \frac{(Z\alpha m_r)^3}{\pi n^3}, \qquad |\psi_{nP}(0)|^2 = 0.$$

So for S states

$$\boxed{\Delta E^{\mathrm{fs}}_{nS} = \frac{2}{3n^3}(Z\alpha)^4 m_r^3 \langle r_E^2\rangle}$$

- $2S$ shifts because it samples the origin.
- $2P$ hardly shifts because $\psi_{2P}(0) = 0$.
- Measured $2S$–$2P$: QED + finite size; fit r_E.

Eides, Grotch & Shelyuto, Phys. Rep. **342**, 63 (2001); Antognini et al., Science **339**, 417 (2013).

Perturbation vs. Non-Perturbative QCD & Lattice (Backup)

Perturbation theory needs a small parameter

$$\mathcal{O} = \mathcal{O}_0 + \alpha_s \mathcal{O}_1 + \alpha_s^2 \mathcal{O}_2 + \cdots .$$

$$\alpha_s \ll 1 \quad \Rightarrow \quad \text{higher orders are suppressed.}$$

Then a few Feynman diagrams can give a controlled approximation.

At proton scales this hierarchy is lost

$$r \sim 1 \text{ fm}, \qquad Q \sim \Lambda_{\mathrm{QCD}} \sim \text{few} \times 100 \text{ MeV}.$$

$$\alpha_s(Q) \sim \mathcal{O}(1) \quad \Rightarrow \quad 1, \alpha_s, \alpha_s^2, \ldots \text{ are not ordered.}$$

Many gluon exchanges, sea-quark fluctuations, and confinement are part of the proton structure, not small corrections.

Therefore the desired current matrix element is non-perturbative

$$\langle p' | J^{\mu}_{\mathrm{em}} | p \rangle \quad \text{must be computed from QCD beyond a low-order diagram expansion.}$$

$$\langle O \rangle = \frac{1}{Z} \int [DU][D\bar{q}][Dq]\, O[U, q, \bar{q}]\, e^{-S_{\mathrm{QCD}}} \quad \xrightarrow[\mathrm{MC}]{\mathrm{lattice}} \quad \text{controlled } a \to 0,\ L \to \infty \text{ result.}$$

Lepage, "Lattice QCD for Novices," arXiv:hep-lat/0506036 (2005); Wittig, "QCD on the Lattice," Springer (2020), Sec. 5.2.

Why Do Gluons Live on Links? (Backup)

Quarks live at spacetime points

$$\psi(x) \to \Omega(x)\,\psi(x), \qquad \bar{\psi}(x) \to \bar{\psi}(x)\Omega^\dagger(x).$$

- $\Omega(x) \in SU(3)$ is a local colour rotation.
- Neighbouring sites have independent colour frames: $\Omega(x) \neq \Omega(x + a\hat{\mu})$.
- A naive hopping term is not gauge invariant:

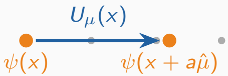
$$\bar{\psi}(x)\psi(x + a\hat{\mu}) \to \bar{\psi}(x)\Omega^\dagger(x)\Omega(x + a\hat{\mu})\psi(x + a\hat{\mu}).$$

The gluon is the connection between sites

$$U_\mu(x) \simeq \exp[iag\, A_\mu(x + a\hat{\mu}/2)] \in SU(3).$$

$$U_\mu(x) \to \Omega(x)U_\mu(x)\Omega^\dagger(x + a\hat{\mu}).$$

$$\bar{\psi}(x)U_\mu(x)\psi(x + a\hat{\mu}) \quad \text{is gauge invariant.}$$



Gauge action: the smallest closed transporter is the plaquette

$$U_{\mu\nu}(x) = U_\mu(x)U_\nu(x + a\hat{\mu})U_\mu^\dagger(x + a\hat{\nu})U_\nu^\dagger(x) \simeq \exp\left[ia^2 g\, F_{\mu\nu}(x)\right].$$

The Wilson plaquette combination obeys

$$1 - \frac{1}{3}\mathrm{Re}\,\mathrm{Tr}\,U_{\mu\nu} \;\propto\; a^4\,\mathrm{Tr}\,F_{\mu\nu}F_{\mu\nu} + \mathcal{O}(a^6),$$

which gives the Euclidean gauge action in the continuum limit.

Why Euclidean, Not Directly Minkowski? (Backup)

Minkowski path integral

$$Z_M = \int [D\Phi]\, e^{iS_M[\Phi]}.$$

- e^{iS_M} is a rapidly oscillating phase.
- It is not a positive probability weight.
- Configurations cancel against each other $\Rightarrow$ severe sign problem.

Wick rotation

$$t = -i\tau, \qquad e^{iS_M} \longrightarrow e^{-S_E}.$$

Euclidean path integral

$$Z_E = \int [D\Phi]\, e^{-S_E[\Phi]}.$$

- e^{-S_E} is a Boltzmann-like weight.
- After integrating out the fermions, zero-density QCD can be sampled by Monte Carlo.
- Euclidean time also filters hadron ground states:

$$C_2(\tau) = \sum_n |A_n|^2 e^{-E_n\tau} \xrightarrow[\tau \text{ large}]{} |A_0|^2 e^{-M_N\tau}.$$

For this talk: the charge radius uses spacelike form factors $G_E(Q^2)$, $Q^2 > 0$, which are naturally accessed from Euclidean correlation functions and then extrapolated to $Q^2 = 0$.

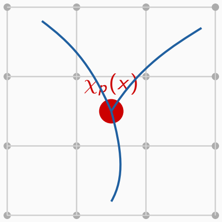
How Is a Proton Put on the Lattice? I (Backup)

Important point: we do not insert a little proton as a particle. The lattice path integral contains quark and gluon fields. A "proton" is created by a **composite operator** with the proton's quantum numbers.

Example: proton interpolating operator

$$\chi_p^\alpha(x) = \epsilon^{abc}\left(u^{aT}(x)C\gamma_5 d^b(x)\right)u^{c\alpha}(x).$$

- uud: proton flavour content.
- ϵ^{abc}: colour singlet.
- $C\gamma_5$: spin-zero diquark structure; parity is selected with a projector.
- α: remaining Dirac spin index.



local product of three quark fields at one site

$$\bar{\chi}_p(0)\left|0\right\rangle = Z_p\left|p\right\rangle + Z_{N^*}\left|N^*\right\rangle + Z_{N\pi}\left|N\pi\right\rangle + \cdots .$$

Meaning: the operator creates every state with proton quantum numbers; Euclidean time later filters out the lightest one, the proton.

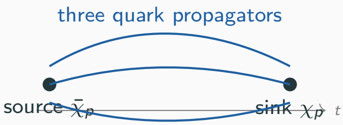
Ioffe, Nucl. Phys. B **188**, 317 (1981); Wittig, "QCD on the Lattice", Springer (2020), §5.2.

How Is a Proton Put on the Lattice? II (Backup)

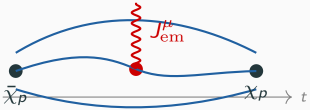
Source and sink. Put $\bar{\chi}_p$ at the source time and χ_p at the sink time; then average over the QCD ensemble.

$$C_2(t) = \sum_{\vec{x}} \langle \chi_p(\vec{x}, t)\, \bar{\chi}_p(\vec{0}, 0)\rangle.$$

$$C_3^{\mu}(t, \tau, \vec{q}) = \sum_{\vec{x},\vec{y}} e^{i\vec{q}\cdot\vec{y}} \langle \chi_p(\vec{x}, t)\, J^{\mu}_{\mathrm{em}}(\vec{y}, \tau)\, \bar{\chi}_p(\vec{0}, 0)\rangle.$$



C_2: create and identify a proton



C_3: insert current and measure response

After integrating out the Grassmann quark fields, these pictures become products of quark propagators $S_q(x, 0) = D_q^{-1}(x, 0)$ on each gauge configuration. Large Euclidean separations isolate the ground-state term; an appropriate C_3/C_2 ratio then removes overlap and propagation factors.

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012).

After Wick Contraction: What Is Computed? (Backup)

Before contraction:

$$C_2(t) = \sum_{\vec{x}} \langle \chi_p(\vec{x}, t)\, \bar{\chi}_p(\vec{0}, 0) \rangle, \qquad \chi_p \sim \epsilon^{abc} (u^{aT} C\gamma_5 d^b) u^c.$$

Fixed gauge configuration U_i: Wick contracting the quark fields gives

$$\langle q(x)\, \bar{q}(y) \rangle_{U_i} = S_q(x, y; U_i) = D_q[U_i]^{-1}(x, y).$$

$\chi_p, \bar{\chi}_p$ fix colour, spin, flavour contraction pattern	→	solve Dirac equation $D_q[U_i] S_q = \mathbf{1}$ to get S_u, S_d	→	multiply/contract propagators then sum over $\vec{x}$

Schematic contracted form:

$$C_2(t; U_i) = \sum_{\vec{x}} \mathcal{C}_{\chi_p} \Big[S_u(x, 0; U_i),\, S_d(x, 0; U_i),\, S_u(x, 0; U_i) \Big],$$

$$C_2(t) \simeq \frac{1}{N_{\rm cfg}} \sum_{i=1}^{N_{\rm cfg}} C_2(t; U_i).$$

$C^{\mu}_{3,\rm conn}$: $\quad S_q(x, 0; U_i) \to S_q(x, y; U_i)\, \Gamma^{\mu}_q\, S_q(y, 0; U_i) \quad \Rightarrow \quad$ appropriate ratio of C^{μ}_3 and $C_2 \to \langle p' | J^{\mu}_{\rm em} | p \rangle$.

Ratio and Summation Methods (Backup)

Ratio method: choose a combination where source/sink overlap factors and leading Euclidean exponentials cancel,

$$R_\mu(\vec{q}, t, \tau) = \frac{C_3^\mu(\vec{q}, t, \tau)}{C_2(\vec{0}, t)} \sqrt{\frac{C_2(\vec{q}, t-\tau)\, C_2(\vec{0}, \tau)\, C_2(\vec{0}, t)}{C_2(\vec{0}, t-\tau)\, C_2(\vec{q}, \tau)\, C_2(\vec{q}, t)}} \xrightarrow{0 \ll \tau \ll t} \Pi_\mu .$$

$$\Pi_\mu(\vec{q}; \Gamma) = K_\mu^E(\vec{q}, \Gamma)\, G_E(Q^2) + K_\mu^M(\vec{q}, \Gamma)\, G_M(Q^2).$$

Known kinematics K_μ: energies, momentum, state normalization, and spin projector Γ.

Summation method: reduce excited-state contamination by summing over current insertion time,

$$S_\mu(\vec{q}, t) = \sum_{\tau=a}^{t-a} R_\mu(\vec{q}, t, \tau) \xrightarrow{t \text{ large}} c + t\,\Pi_\mu + \mathcal{O}(e^{-\Delta E\, t}).$$

Signal-to-noise

- baryon noise worsens exponentially with t
- use large statistics, smearing, multiple source-sink separations

Excited states

- N^*, $N\pi$, $N\pi\pi \ldots$ also contribute
- use summation, multi-state fits, or GEVP

Maiani et al., NPB **293**, 420 (1987); Capitani et al., PRD **86**, 074502 (2012); Lepage (1989).

Why Disconnected Diagrams Matter (Backup)

Full QCD current insertion has two topologies.

$$J^\mu_{\mathrm{em}} = \sum_q e_q\, \bar{q}\gamma^\mu q \quad \Rightarrow \quad C_3^{\mathrm{conn}} + C_3^{\mathrm{disc}}.$$

Connected

- J^μ_{em} hits a quark line joining source and sink.
- Physical picture: direct response of the proton quark lines.

Disconnected

- J^μ_{em} hits a sea-quark loop.
- Not connected by a quark line, but correlated through the same gluon background U.
- This is a genuine sea-quark fluctuation, not an IR divergence.

What is computed on each gauge field U_i?

$$L^\mu_q(t; U_i) = \sum_{\vec{x}} \mathrm{Tr}_{c,s}\left[\gamma^\mu\, D_q^{-1}(x, x; U_i)\right]$$

$$C^\mu_{3,\mathrm{disc}}(t, \tau) \sim \left\langle C_2(t; U)\, L^\mu_q(\tau; U)\right\rangle_U - \left\langle C_2(t; U)\right\rangle_U \left\langle L^\mu_q(\tau; U)\right\rangle_U.$$

- $D_q^{-1}(x, x)$: all-to-all quark propagation back to the same point.
- Trace over colour and spin $\Rightarrow$ stochastic estimators.
- Numerically noisy and expensive, but required for precision form factors.

Djukanovic et al., PRD **109**, 094510 (2024); Djukanovic et al., PRL **132**, 211901 (2024).

Flavor Currents: What Are $u - d$ and $u + d$? (Backup)

Start from quark-flavor vector currents

$$J_u^\mu = \bar{u}\gamma^\mu u, \qquad J_d^\mu = \bar{d}\gamma^\mu d, \qquad J_s^\mu = \bar{s}\gamma^\mu s.$$

Isovector: $u - d$

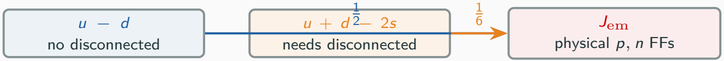
$$J_{u-d}^\mu = J_u^\mu - J_d^\mu.$$

- Measures the proton–neutron difference:
$$G^{u-d} = G^p - G^n.$$
- In the isospin limit $m_u = m_d$, disconnected loops cancel:
$$L_u^\mu - L_d^\mu = 0.$$
- Hence $u - d$ is the cleanest lattice channel.

Isoscalar / octet: $u + d$, $u + d - 2s$

$$J_{u+d}^\mu = J_u^\mu + J_d^\mu, \qquad J_8^\mu = J_u^\mu + J_d^\mu - 2J_s^\mu.$$

- Measures the proton–neutron common part.
- Disconnected loops add instead of cancel:
$$L_u^\mu + L_d^\mu \neq 0.$$
- Full $N_f = 2 + 1$ EM current uses the octet combination:
$$J_{\mathrm{em}}^\mu = \frac{1}{2} J_{u-d}^\mu + \frac{1}{6} J_8^\mu.$$



Djukanovic et al., PRL **132**, 211901 (2024), lattice setup and flavor decomposition.

How to Read $u-d$ and $+\mathrm{exp.}(n)$ (Backup)

The labels in PRL Fig. 3 do not all mean the same level of calculation.

$u-d$

$$G^{u-d} = G^p - G^n.$$

- Pure isovector lattice result.
- Useful because disconnected diagrams cancel for $m_u = m_d$.
- It is not yet a full proton EM result; it gives the proton–neutron difference.

$+\mathrm{exp.}(n)$

$$G^p = G^{u-d} + G^n_{\mathrm{exp}} \quad \text{schematically.}$$

- Combine lattice $u-d$ with experimental neutron input.
- Good comparison, but not a fully ab initio proton result.
- For charge radii, the slope combination gives schematically
$$\langle r_E^2 \rangle^p = \langle r_E^2 \rangle^{u-d} + \langle r_E^2 \rangle^n_{\mathrm{exp}}.$$

Speaking sentence: $u-d$ is clean but incomplete; $+\mathrm{exp.}(n)$ uses neutron data to reconstruct a proton number.

PRL Fig. 3 caption: Mainz21 combines isovector results with PDG neutron values.

How to Read “This Work” in PRL Fig. 3 (Backup)

This work

$$J^{\mu}_{\mathrm{em}} = \frac{1}{2}(u-d) + \frac{1}{6}(u+d-2s).$$

- $u-d$ part: connected isovector channel.
- $u+d-2s$ part: isoscalar/octet channel.
- Disconnected sea-quark loops are included instead of assumed away.
- Continuum, finite-volume, and physical-point extrapolations are controlled.
- Therefore this is the final first-principles proton result:

$$\sqrt{\langle r_E^2\rangle^p} = 0.820(14)\ \mathrm{fm}.$$

$u - d$
difference only → $+\mathrm{exp.}(n)$
mix lattice+data → This work
full EM

PRL Fig. 3 caption: only ETMC19 and this work include disconnected contributions; Djukanovic et al., PRL **132**, 211901 (2024).

The z-Expansion: Analytic Radius Extraction (Backup)

Problem: lattice data at discrete $Q^2 > 0$; need $dG_E/dQ^2|_{Q^2=0}$.
Traditional: dipole $G_E(Q^2) = (1 + Q^2/M_D^2)^{-2}$ — model-dependent.

z-**expansion** (analyticity-based parametrisation):

1. Conformal map $Q^2 \to z$: $\quad z = \frac{\sqrt{t_{\rm cut}+Q^2}-\sqrt{t_{\rm cut}-t_0}}{\sqrt{t_{\rm cut}+Q^2}+\sqrt{t_{\rm cut}-t_0}}$
   $t_{\rm cut} = 4m_\pi^2$ (isovector) or $9m_\pi^2$ (isoscalar); t_0: free param, optimises convergence
2. $|z| < 1$ guaranteed $\to$ convergent Taylor series
3. Expand: $G_E(z) = \sum_{k=0}^{k_{\rm max}} a_k\, z^k$; coefficient priors/bounds and truncation tests control higher orders

For the proton, $G_E^p(0) = 1$, and the radius is

$$\langle r_E^2 \rangle^p = -6\, \frac{dG_E^p}{dz}\, \frac{dz}{dQ^2}\bigg|_{Q^2=0}.$$

If $t_0 = 0$, then $z(0) = 0$ and only a_1 enters the derivative. In Djukanovic et al., the z-expansion is a cross-check; the quoted central result uses model-averaged direct BχPT fits.

Hill, Paz, PRD **82**, 113005 (2010); Hill, EPJ Web Conf. **137**, 01023 (2017).

Finite Volume: How Large L is Needed? (Backup)

Periodic spatial box: $\vec{x} \equiv \vec{x} + L\hat{e}_i$ gives mirror images and quantised momenta. For many stable one-hadron observables, leading finite-volume effects scale as $\mathcal{O}(e^{-m_\pi L})$.

- **Empirical heuristic**: often $m_\pi L \gtrsim 4$ for masses & single-hadron overlaps $\Rightarrow$ tail $\sim e^{-m_\pi L}$ small at chosen tolerance (still observable-dependent).
- The safe procedure is to vary L or use finite-volume formulae; there is no universal box-size criterion for every observable.
- CLS open temporal boundaries address topology freezing at fine a; they do *not* remove spatial finite-volume effects.

Lüscher, Commun. Math. Phys. **104**, 177 (1986); Commun. Math. Phys. **105**, 153 (1986); Beane et al., Nucl. Phys. B793, 379 (2008).

RG and Symanzik Scaling (Backup)

RG continuum limit: changing a means retuning bare parameters $g_0(a), m_q(a)$ along a line of constant physics. Long-distance dimensionless observables are held fixed while $a \to 0$.

$$\mu \frac{dg_R}{d\mu} = \beta(g_R), \quad \mu \sim a^{-1}, \qquad r_E(a) = r_E^{\mathrm{cont}} + c_a a^2 + \cdots .$$

Symanzik effective theory: a lattice observable is described by continuum QCD plus cutoff corrections organised in powers of a. After non-perturbative $O(a)$ improvement of the action and current, the leading fit ansatz used here is a^2, up to logarithms and higher powers.

Therefore: values at several lattice spacings test the predicted scaling and determine the intercept at $a = 0$. This is a controlled continuum extrapolation, not a free-form curve fit.

Symanzik, Nucl. Phys. **B226**, 205 (1983); Bruno et al., JHEP **02**, 043 (2015).

Sea Quarks and Gauge Configurations (Backup)

Start with quark and gluon fields

$$Z = \int [DU][D\bar{\psi}D\psi] \exp\left[-S_g[U;\beta] - \sum_f \bar{\psi}_f D_f[U;m_f]\psi_f\right].$$

Integrate Grassmann fermions analytically

$$Z = \int [DU] \underbrace{e^{-S_g[U;\beta]} \prod_f \det D_f[U;m_f]}_{P[U] \text{ up to normalization}}.$$

What is sampled?

- The generated configurations are $U_\mu(x)$, not $\psi(x)$.
- The probability of U includes sea-quark effects:
  $$P[U] \propto e^{-S_g[U;\beta]} \prod_{f=u,d,s,\ldots} \det D_f[U;m_f].$$
- In HMC the determinant is represented by pseudofermion fields:
  $$\det(D^\dagger D) \propto \int [D\phi^\dagger D\phi]\, e^{-\phi^\dagger (D^\dagger D)^{-1}\phi}.$$

Valence quark lines

- After choosing U_i, solve a Dirac equation:
  $$D_q[U_i;m_q]\, S_q[U_i;m_q] = \mathbf{1}.$$
- $S_q = D_q^{-1}$ is the quark propagator used in C_2, C_3.
- Fermions enter twice: determinant for sea quarks; propagators for measured quark lines.

Where Does m_π Enter the Gauge Weight? (Backup)

m_π is not put into the path integral directly. It is the pion mass produced by the chosen light-quark masses in the fermion determinant.

Example: $2+1$-flavour gauge weight

$$P[U] \propto e^{-S_g[U;\beta]} \det D_l[U;m_l]^2$$
$$\times \det D_s[U;m_s], \qquad m_l \simeq m_u \simeq m_d.$$

- m_l changes the determinant, so it changes the ensemble.
- The generated U_i's are distributed according to that choice of (β, m_l, m_s).
- A lighter m_l produces a lighter *pion*.

Chiral intuition

$$m_\pi^2 \simeq 2B\, m_l \quad (m_l = (m_u + m_d)/2).$$

- Modern ensembles tune m_l so that $m_\pi \simeq m_\pi^{\mathrm{phys}}$.
- Older/heavier ensembles extrapolate observables in m_π.
- For a unitary calculation, valence and sea masses match; otherwise it is partially quenched.

Doublers and the Wilson Operator (Backup)

Naive lattice fermions double the species

$$D_{\mathrm{naive}}(p) = m + \frac{i}{a}\sum_{\mu}\gamma_\mu \sin(ap_\mu).$$

Since $\sin(ap_\mu) = 0$ at $p_\mu = 0$ and $p_\mu = \pi/a$, one flavour becomes $2^4 = 16$ light fermions in four dimensions.

Wilson's cure: add a momentum-dependent mass for the doublers,

$$D_W(p) = D_{\mathrm{naive}}(p) + \frac{r}{a}\sum_{\mu}\big(1 - \cos(ap_\mu)\big).$$

For $p_\mu \sim \pi/a$, this mass is $\mathcal{O}(1/a)$, so doublers decouple as $a \to 0$.

Coordinate-space Wilson operator

$$(D_W q)(x) = \left(m_0 + \frac{4r}{a}\right) q(x) - \frac{1}{2a}\sum_{\mu}\Big[(r-\gamma_\mu)U_\mu(x)q(x+\hat{\mu}) + (r+\gamma_\mu)U^\dagger_\mu(x-\hat{\mu})q(x-\hat{\mu})\Big].$$

$$D_{\mathrm{SW}} = D_W + c_{\mathrm{SW}}\frac{ia}{4}\sigma_{\mu\nu}\widehat{F}_{\mu\nu} \qquad \text{(clover improvement).}$$

- Wilson term removes doublers but breaks chiral symmetry at finite a.
- Clover improvement removes leading $\mathcal{O}(a)$ artefacts after tuning.
- Other fermion actions make different compromises; see next backup slide.

Common Fermion Discretisations (Backup)

The issue: doublers and chiral symmetry cannot both be handled for free. Different lattice fermions make different compromises.

Wilson / clover

- removes doublers with Wilson term
- chiral symmetry broken at $a > 0$
- clover term improves $\mathcal{O}(a)$ artefacts
- moderate cost; good for nucleon structure

Twisted-mass Wilson

- Wilson-type action at nonzero twist
- automatic $\mathcal{O}(a)$ improvement at maximal twist
- often used for nucleon EM observables

Staggered / HISQ

- cheap; remnant chiral symmetry
- has taste degrees of freedom
- common in HVP, thermodynamics, heavy flavour

Domain-wall / overlap

- near-exact / Ginsparg–Wilson chiral symmetry
- expensive
- useful for kaon physics and chiral observables

This talk: CLS $N_f = 2 + 1$ Wilson–clover ensembles — $\mathcal{O}(a)$-improved, moderate cost, chiral symmetry restored as $a \to 0$.

Nielsen & Ninomiya (1981); Sheikholeslami & Wohlert (1985); Frezzotti & Rossi (2004); Follana et al. (2007); Kaplan (1992); Neuberger (1998).

Zemach and Friar Radii (Backup)

Beyond r_E, the **Friar radius** (r_F) enters higher-order finite-size corrections to the Lamb shift, while the **Zemach radius** (r_Z) enters hyperfine splitting.
($\kappa_p = \mu_p/\mu_N - 1 \approx 1.793$: proton anomalous magnetic moment)

Definitions (convolutions of form factors):

$$r_Z = -\frac{4}{\pi}\int_0^\infty \frac{dQ}{Q^2}\left[\frac{G_E G_M}{1+\kappa_p} - 1\right], \quad r_F^3 = \frac{48}{\pi}\int_0^\infty \frac{dQ}{Q^4}\left[G_E^2 - 1 + \tfrac{1}{3}\langle r_E^2\rangle Q^2\right]$$

First lattice QCD calculation (Mainz group):

$$r_Z^p = 1.013(10)(12)\text{ fm}, \qquad r_F^p = 1.301(12)(14)\text{ fm}$$

- r_F: higher-order charge-distribution correction in muonic atoms
- r_Z: leading elastic proton-structure correction to hyperfine splitting
- Compatible with most phenomenological determinations
- Computed from the same CLS ensembles and analysis framework

Djukanovic, von Hippel, Meyer, Ottnad, Salg, Wittig, PRD **110**, L011503 (2024) [arXiv:2309.17232].

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